

Supervisor Meeting — Progress & Discussion

Notes

IKS-Grounded Multimodal Agricultural Advisory System M.Tech Thesis · IIITDM Jabalpur · Ankit Pawar · Supervisor: Dr. Akshay Pandey Meeting date: 25 August 2026

Purpose of this document. *A single, up-to-date briefing for today's meeting. It summarises the complete system as it stands, the latest accuracy figures for every component, and — importantly — the **evaluation results and the disease-model redesign that were completed since our last meeting** and have not yet been reviewed together. The final two sections list what remains and the specific points to discuss.*

1. Executive summary

The system runs **end-to-end and live**. A farmer uploads a leaf photo and a soil photo, declares the crop, and receives advice that is **grounded in and cited from classical Indian agricultural texts** (Vrikshayurveda, Brihat Samhita, Krishi Parashara, Upavanavinoda) — or an **honest refusal** when the texts do not cover the case.

Since the last meeting, three substantial pieces of work were completed:

1. **A full evaluation phase** — retrieval quality, answer grounding, and a RAGAS faithfulness assessment (new; not yet reviewed together).
2. **A redesign of how the texts are used** — from crop-specific lookup to **symptom-based retrieval**, matching how the classical texts actually prescribe treatment, plus **calibrated-confidence handling for untrained plants** instead of a hard refusal.
3. **A crop-agnostic "disease-type" classifier experiment** — an isolated study (the deployed model is untouched) that directly supports the cross-plant generalisation idea.

The scientific centre of gravity remains the **disease model's honesty problem** — the earlier concern that the model attended to the background rather than the leaf. That has been diagnosed and fixed, and the fix now verifiably holds even on field images.

2. The two disease models — a simple comparison

This is the main thing to decide today. It can be explained in two minutes.

We now have **two** disease models. They are **not** better-or-worse versions of the same thing — they answer **two different questions**.

Model 1 — "C-PD" (the one running in the system now)

- **What it tells you:** the crop *and* the disease together — for example, "Tomato — Late blight".
- **How many classes:** 27 (each class is one crop + one disease).

- **How it was built:** started from our clean lab images (PlantVillage), then re-trained on **cropped leaf photos** from real-world images (PlantDoc). It only knows the crops in those datasets.
- **Accuracy:** 66.6%, and it honestly looks at the leaf.
- **Its limit:** it only recognises crops it was trained on. Show it a plant it never saw, and it has no matching label for it.

Model 2 — the symptom-based "disease-type" model (the new experiment)

- **What it tells you:** only the **disease type**, without the crop — for example, just "Blight" (or "Rust", "Leaf spot").
- **How many classes:** about 13 (each class is one disease type; the crop name is removed).
- **How it was built:** same lab starting point, then trained on a **mix of datasets** (PlantVillage + PlantDoc + the Brazilian multi-crop set), all re-labelled **by disease type only**, on cropped leaves.
- **Accuracy:** 71.9% (the leaf-cropped version).
- **Its strength:** because the crop is removed, it can recognise a disease on **any plant — even one it never saw**. This matches our classical texts, which prescribe by **symptom, not by crop**.

Side-by-side

	Model 1 — C-PD (current)	Model 2 — Symptom-based (new)
What it names	Crop and disease	Disease type only
Example answer	"Tomato — Late blight"	"Blight"
Number of classes	27 (crop + disease)	~13 (disease type only)
Trained on	PlantVillage → cropped PlantDoc	PlantVillage + PlantDoc + Brazilian mix
Works on an unseen crop?	❌ No	✅ Yes
Matches the IKS texts (symptom-based)?	Partly	✅ Directly
Accuracy	66.6%	71.9%
Status	✅ Proven, running in the system	🔬 New experiment, not yet deployed

(Note: the two accuracy numbers are not a fair head-to-head — Model 2 has fewer, broader classes, so its number is naturally a little higher. They measure different tasks.)

The difference in one line: *C-PD tells you which crop has which disease; the symptom model tells you which disease it is, on any crop.*

Why the new model fits our thesis novelty: our classical texts treat by symptom ("white spot"), not by crop name. A model that thinks in **disease types** — not crops — matches the texts and lets us help farmers even with crops we never trained on.

My recommendation: keep C-PD running (it is proven and ready to demo), and **present the symptom model as our novelty and future direction**. Only swap it in later, after one more test (training it while hiding one crop, then testing on that crop) proves it truly works on unseen plants.

3. System at a glance

Three model families, wired together by an integration layer:

- 1. **Disease classifier** — EfficientNet-B4. Leaf photo → disease prediction, with a Grad-CAM heatmap showing where the model looked. A pretrained leaf detector crops the leaf **before** the classifier sees it, so the background cannot act as a shortcut.
- 2. **Soil classifier** — EfficientNet-B0, multi-task (soil type, moisture, texture).
- 3. **RAG advisory** — hybrid retrieval (keyword + semantic + re-ranking) over a ChromaDB of the classical-text corpus, with grounded generation by Llama-3.1-8B (frozen; used for retrieval-and-generation only, never fine-tuned).
- 4. **Integration** — converts the vision predictions into a **symptom-led query** in the vocabulary the classical texts actually use.
- 5. **Explainability** — Grad-CAM for disease and soil; highlighting of the exact retrieved passages used.
- 6. **Interface** — a working Streamlit application.

4. Component results (current numbers)

4.1 Disease classifier

The classifier is a three-stage transfer cascade. Its behaviour, stage by stage:

Stage	Data	Accuracy	Where the model looks
Clean lab images (PlantVillage)	38 classes	99.8%	leaf ✓
Field canopy (Paddy)	10 classes	97.0%	lesion ✓
In-the-wild (PlantDoc)	27 classes	72.3%	background ✗ (the honesty problem)

For published context, the best PlantDoc results in the literature are ~73–78%, so **72.3% is at the research frontier** — the concern was never the number, but *how* the model reached it (by reading the background).

The fix that worked — retrain on leaf crops ("C-PD"). Retraining on cropped leaf regions, warm-started from the general PlantVillage backbone, produces a model at **66.6% top-1 with honest, leaf-focused attention** (Grad-CAM now sits on the leaf, not the background). The guiding principle we adopted: *a correct model at 66.6% is worth more than a background-reading model at 72.3%.*

An important reframe of the 66.6%. Healthy-vs-diseased detection is near-perfect. Most of the remaining errors are confusions *between disease subtypes of the same crop* (e.g. corn rust vs corn blight), not healthy/diseased mistakes. So the model reliably answers the question that matters to a farmer — *does this plant need attention?* — even when it debates the exact subtype.

4.2 Soil classifier (production model)

Property	Top-1 accuracy
Soil type	89.9%
Moisture	95.8%
Texture	67.9%

Texture is the weakest head and remains a genuinely hard problem — a candidate for future work rather than a quick win.

4.3 Retrieval and integration

The key design choice is how a modern vision label becomes a query against classical text. A direct template ("Apple Scab Leaf") barely retrieves anything (score 0.01–0.04), because the texts never use that vocabulary.

Rewriting the label into a symptom description the texts actually use ("scorched leaves with whitish spots") lifts retrieval quality to **0.59–0.96**. This bridge is the core integration contribution.

4.4 Grounded generation

The generator answers **only** from the retrieved passages and cites each one by source, chapter, and verse. When the corpus does not cover a case, it refuses rather than inventing an answer.

5. New since our last meeting

These three items were completed after our previous discussion and are the main things to review together today.

5.1 Evaluation phase — first full results

We ran a structured evaluation over a set of 24 symptom-led queries (22 answerable + 2 that should be refused).

Retrieval quality (higher is better; comparing the full system to a keyword-only baseline):

Configuration	nDCG@5	MRR	Precision@5	Hit@5
Full (semantic + keyword + re-rank)	0.94	0.91	0.74	1.00
Keyword-only (baseline)	0.70	0.62	0.56	0.91
Semantic-only	0.94	0.94	0.80	1.00

Answer grounding and honesty:

- **Honest refusal on the negatives: 100%** — it correctly declines the two questions the corpus cannot answer.
- **Fabricated citations: 0%** — it never cites a source it did not use.
- **Over-refusal: ~55%** — on answerable questions, it refuses about half the time.
- **Faithfulness (RAGAS, when it does answer): 0.56**; an independent cross-check on the answered subset put faithfulness at **0.86**.

The headline finding. The system is *safe and faithful* — when it answers, the answer is grounded, and it never fabricates. **The limiter is corpus coverage, not the retrieval or the model.** The ~206 tree-focused passages simply do not cover every symptom a farmer might present, so the generator honestly refuses rather than guessing. This directly motivates expanding the corpus (discussed in §8).

5.2 Symptom-based retrieval + untrained-plant handling

Two design changes, both aligned with how the classical texts actually work:

- **Symptom-based, not crop-specific.** The texts prescribe by general symptom (e.g. "white spot"), not by a named crop. The retrieval was realigned to **lead with the symptom** and treat the crop as background context, rather than refusing when a specific crop is not named in the text.
- **Untrained plants — advise with calibrated confidence, not a flat refusal.** Because a disease's visual appearance transfers across species, the classifier's disease knowledge is useful even on a plant it was never trained on. Instead of refusing, the system now runs the model, shows a **calibrated confidence percentage** (using temperature scaling, a standard technique that corrects an over-confident model without changing its predictions), and advises via the symptom-based retrieval **with a clear caution**. Only when confidence is genuinely low does it hold back, collect the image, and ask the farmer for more photos. Wording is deliberately honest — "confidence", never "accuracy".

5.3 Crop-agnostic "disease-type" experiment

An **isolated research experiment** (the deployed model, its data, and its configuration were left completely untouched; everything is reversible). The idea: pool the many crop-specific disease labels into ~13 **crop-agnostic disease types** (rust, blight, leaf spot, and so on) — directly aligned with the symptom-based direction and with the untrained-plant idea.

Version	Test accuracy	Attention on field images
Without leaf-cropping	78.2%	reads the background ✖
With leaf-cropping	71.9%	on the leaf/lesions ✔

The -6 accuracy points are the honest cost; the leaf-focused attention is what they buy. Grad-CAM on the cropped model confirmed the fix holds on the hardest cases — corn-in-a-field images now heat the lesions, with the soil going cold. This is the same trade-off, and the same verdict, as the main disease model: **keep the leaf-focused version**. The weak classes (a couple with very few samples, and an early-blight/late-blight confusion) are known and fixable.

5.4 On the datasets provided

The disease dataset provided earlier was inspected carefully. It is a re-packaged copy of **PlantVillage** — the same lab-style images the project already uses (confirmed by an exact class-count match and the shared naming convention), with **no field-style imagery**. Merging it would add no new information and would risk re-amplifying the very lab-only shortcut bias the project has been working to remove. The one genuinely useful piece is its "background / no-leaf" folder, usable as a guardrail when a farmer uploads a non-leaf photo.

This is a specific request for the meeting (see §9): what would help most is **field-style imagery** — plot photographs with soil and neighbouring plants visible — as that is exactly what reduces the background-shortcut bias.

6. What worked, and what did not (kept honest)

Documented negative results strengthen the thesis:

- **Background randomisation** (compositing leaves onto random backgrounds): cost accuracy for almost no attention gain. Abandoned.
- **Linear-probe fine-tuning** on a frozen rice-specialised backbone: dropped to 61%. Abandoned.
- **Cropping at inference only** (without retraining): dropped to 58.2% — proof that a background-trained model cannot be fixed by cropping alone; it must be **retrained** on crops. This is *why* the C-PD retrain was necessary.
- **Sequential soil transfer**: training collapsed. Abandoned in favour of the stable production recipe.

The consistent lesson: **the leaf-focus problem is only solved by retraining on leaf crops**, which is exactly what the deployed model now does.

7. Where the project stands

Component	Status
Classical-text corpus + retrieval	✓ Working
Grounded generation with citations	✓ Working
Soil classifier	✓ Production
Disease classifier (honest, leaf-focused)	✓ Working
Symptom-based retrieval + untrained-plant handling	✓ Implemented
Full-system interface	✓ Working (live demo)
Evaluation phase	✓ First full pass complete
Corpus coverage expansion	⌚ Open — the main limiter
Expert (gold) evaluation set	⌚ Open — needs expert judgement

8. What remains

1. **Expand the corpus** — the evaluation shows coverage, not method, is the limiter. Adding a small number of well-chosen texts would directly reduce the ~55% over-refusal.
2. **Build an expert gold-standard evaluation set** — the current results use silver (approximate) labels; a small expert-judged set would make the evaluation citable.
3. **Finish the disease-type follow-ups** — merge the early/late blight classes, address the tiny under-sampled classes, and run a **held-out-crop test** (train while excluding one crop, then test on it) — the strongest single piece of evidence for the cross-plant generalisation claim.
4. **Continue the crop/soil suitability reference list** — the list of 120 crops with soil suitability is in progress.

9. Points to discuss today

1. **Corpus expansion.** The evaluation shows the system is faithful but limited by how much the texts cover. **Which additional classical or authoritative texts would be appropriate to add**, and are there any specific sources recommended?
2. **Field-style imagery.** Would field photographs (soil + neighbouring plants visible) be available? That is the single most valuable data addition for reducing the background bias — far more than more lab images.
3. **Symptom-based treatment — validation.** The retrieval now treats the texts as symptom-based rather than crop-specific. **Is this the correct reading of the classical treatment logic?** Confirmation here validates a core design decision.
4. **Untrained-plant advice.** The system now advises on plants it was not trained on, using a calibrated confidence figure and an explicit caution, rather than refusing. **Is this the right level of caution** for a farmer-facing tool, and is the confidence-plus-caution framing acceptable?
5. **Expert evaluation.** For a citable evaluation, a small set of expert-judged answers is needed. **Could a short set of queries be reviewed for correctness**, or a suitable expert suggested?
6. **The disease-model accuracy trade-off.** The honest, leaf-focused model scores a few points lower than the background-reading one. **Confirmation that trading a small amount of headline accuracy for genuine, explainable leaf attention is the right call for the thesis.**
7. **Crop-agnostic disease types.** The crop-agnostic direction is promising and aligns with the symptom-based logic. **Is it worth developing further** as a thesis contribution, or kept as a supporting experiment?

Prepared for the supervisor meeting on 25 August 2026. All figures are drawn from the project's experiment log; nothing here is projected or estimated.